

Dr. Ganesh Kumar M

B.Tech, M.Tech, Ph.D.

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Career Objective:

To be associated with a Progressive Organization that gives scope to apply my knowledge and Skills, and to be a part of a team that dynamically works towards the growth of the Organization.

Academic Qualifications:

Degree	Year of completion	University	Institute	Specialisation	Marks/ CGPA
Ph.D. (Internal Full Time Category) Admitted in 2020	2024	VIT-AP University	VIT-AP University, School of Electronics Engineering	Image Processing (using Deep Learning techniques)	9 CGPA
M.Tech	2013	JNTU Anantapur	Geethanjali College of Engg & Tech; Kurnool	VLSI System Design	76.64%
B.Tech	2011	JNTU Anantapur	SVR Engineering College, Nandyal	Electronics & Communication Engineering	75.98%
Intermediate	2007	Board of Intermediate Education, A.P.	Vijayawada Nalanda Junior College, Nandyal	Maths, Physics, Chemistry	88.7%
SSC	2005	Board of Secondary Education, A.P.	Good Shepherd English Medium School, Nandyal	English Medium	82.16%

Subjects Taught:

Basic Electronics, Digital Electronics, Radar Systems, Digital Image Processing, Communication systems, Electronic Measurements and Instrumentation, Electronic Devices and Circuits, Cellular and Mobile Communications, Software Engineering, Internet of Things Architectures and Protocols.

Interested Subjects:

Internet of Things, Digital Image Processing, Basic Electronics, Digital Electronics, Radar Systems Quantum Computing, Python, Artificial Intelligence, Machine Learning, Deep Learning, etc.

Ph.D. Work:

Ph.D. Thesis Title:

Knee Osteoarthritis Detection and Severity Classification Using Deep Learning Classifiers: Advancing Automated Diagnostic Solutions in Healthcare.

Summary on Ph.D. Work: KOA detection and classification are performed in this research using three different classifiers such as DCNN, Social wolf swarm optimized DCNN, and MSO-tuned DCNN classifier. KOA must be identified in order to make a diagnosis and comprehend the severity of a patient's disease. Medical personnel can better manage symptoms, enhance joint performance, and improve patients' standard of living by understanding KOA. By examining features and patterns that the human eye might not immediately detect, DCNN classifiers have proven to be highly accurate at classifying KOA. This could decrease human error and boost diagnostic precision. The utilization of different optimization algorithms also helped in boosting the convergence. In conclusion, classifying KOA using a DCNN classifier has a number of potential advantages and real-world uses. By combining the capabilities of deep learning and image processing, a CNN classifier can aid in the precise and automatic identification of KOA based on imaging information from X-rays or MRIs.

Future work: Continued research into ML and AI approaches can improve KOA diagnosis, prognosis, and treatment. This includes creating increasingly complex AI image analysis algorithms, using ML to forecast the course of diseases, and incorporating AI-driven decision support systems into clinical practice.

Summary of Publications:

Type of Publications / Academic Year	2021-22	2022-23	2023-24	2024-25	2025-26	Total
Funded Project	Nil	Nil	Nil	Nil	UTP-10,000 USD	01
SCI / ESCI / SCIE Journals	Nil	02	01	01	01	05
Scopus Journals	Nil	01	Nil	02	Nil	03
IEEE / JOP / AIP Conferences	01	Nil	Nil	03	04	08
Patents	Nil	04	Nil	07	10	21 (07 Granted, 14 Published)
Book Chapters	Nil	Nil	Nil	03 (Accepted)	01 (Taylor & Francis)	04
Text Book	Nil	Nil	Nil	01 Published	02 Published	03
Grand Total						44

Funded Project:

1. **Principal Investigator** in QIS-UTP Matching Grant Project titled” AI Technology for A Patient Centered Digital Transformation to improve healthcare information system in Government Hospital” a funded project worth **10,000 USD** between QIS College of Engineering and Technology and Universiti Teknologi PETRONAS, Seri Iskandar Malaysia.

Journals:

1. M. Ganesh Kumar, L. Bharathi, M. Senthil, P. Koteswara Rao, G. Rajya Lakshmi Knee osteoarthritis classification using Optimized Granger Causality Inspired Graph Neural Network with Accelerated Model-agnostic Explanations, The Knee, Volume 61, 2026, 104365, ISSN 0968-0160, <https://doi.org/10.1016/j.knee.2026.104365>. **(SCIE Indexed), Q2, Impact Factor 2.0.**
2. M. Ganesh Kumar, Sreenath Kocharla, Narikamalli Yaswanth, T. Vijaya Narashimha Swamy, U. Prasad, T. Vamsee, EHA-LNN: Optimized light gradient-boosting machine enabled neural network for cancer detection using mammography, Biomedical Signal Processing and Control, Volume 105, 2025, 107540, ISSN 1746-8094, <https://doi.org/10.1016/j.bspc.2025.107540>. (<https://www.sciencedirect.com/science/article/pii/S1746809425000515>). **(SCIE Indexed), Q1, Impact Factor 4.9.**
3. Kumar, Ganesh and Gumma, Lakshminarayana and Neelam, Saikiran and Yaswanth, Narikamalli and yedukondalu, Jammisetty (2024). Early Knee Osteoarthritis Classification Using Distributed Explainable Convolutional Neural Network with Local Interpretable Model-Agnostic Explanations. Engineering Research Express, {<http://iopscience.iop.org/article/10.1088/2631-8695/ad9b6f>} **(Scopus Indexed).**
4. Ganesh Kumar M & Agam Das Goswami (2024), Effective Classification of Knee Osteoarthritis using Military scrutoif optimization tuned Deep Convolutional Neural Network Classifier, Biomedical Engineering: Applications, Basis and Communications (BME), <https://doi.org/10.4015/S1016237224500108>. **(ESCI, Scopus Indexed).**
5. Bhavekar, G., Chafle, P., Goswami, A., Marathula, G., Hirve, S., Karpe, S., Magar, N., Farakte, A., Pikle, N., Shinde, S., & Gaikwad, A. (2024). Hybrid approach to medical decision-making: prediction of heart disease with artificial neural network. Bulletin of Electrical Engineering and Informatics, 13(6), 4124-4133. doi:<https://doi.org/10.11591/eei.v13i6.5583> **(Scopus Indexed).**

6. Ganesh Kumar M & Agam Das Goswami (2023) Knee osteoarthritis classification using social wolf swarm-based deep convolutional neural network, "Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization", DOI: [10.1080/21681163.2023.2203266](https://doi.org/10.1080/21681163.2023.2203266) **Taylor & Francis. (ESCI, Scopus Indexed).**
7. Kumar M, G. ., & Goswami, A. D. . (2023). Deep Convolutional Neural Network Classifier for Effective Knee Osteoarthritis Classification . "International Journal on Recent and Innovation Trends in Computing and Communication", 11(3), 242–249.
<https://doi.org/10.17762/ijritcc.v11i3.6343>. **(Scopus Indexed).**
8. M GK, Goswami AD. Automatic Classification of the Severity of Knee Osteoarthritis Using Enhanced Image Sharpening and CNN. *Applied Sciences*. 2023; 13(3):1658.
<https://doi.org/10.3390/app13031658> **(SCIE Indexed).**

Conferences:

1. **M. G. Kumar**, I. Shalout, D. Chaitanya, S. Kumar, S. S. Jennifer and D. F. Sanjaybhai, "An Effective Multidrug-Resistant Tuberculosis Detection Using DenseNet121 Feature Treatment and Feed-Forward MLP Classification," *2026 International Conference on Electronic Systems and Intelligent Computing (ICESIC)*, Chennai, India, 2026, pp. 858-863, doi: 10.1109/ICESIC67389.2026.11496396. **(IEEE Conference)**
2. S. M, **M. G. Kumar**, S. R. Basha, S. Padma, K. TS and D. K. P, "An Improved Power Management Solutions for DC Microgrids Featuring Electric Vehicle Fast Charging Stations," *2025 Tenth International Conference on Science Technology Engineering and Mathematics (ICONSTEM)*, Chennai, India, 2025, pp. 1-9, doi: 10.1109/ICONSTEM65670.2025.11374550. **(IEEE Conference)**
3. **M. G. Kumar**, A. A. A. Samhan, N. S. R. Srikanth, V. D. Babu, R. Bhookya and S. B, "Efficient Cost Evaluation and Hybrid Optimization-Based Heterogeneous Resource Allocation in Cloud–Edge-IoT Environment," *2025 2nd International Conference on Artificial Intelligence and Knowledge Discovery in Concurrent Engineering (ICECONF)*, Chennai, India, 2025, pp. 1-10, doi: 10.1109/ICECONF65644.2025.11379589. **(IEEE Conference)**
4. **G. K. Marathula**, I. Venu Gopal, AmnehAlamleh, M. Gopila, T. Nagini and S.Sivakami, "An Efficient Methodology to Predict Train Delays using Intelligent Hybrid Deep Learning Procedure," *2025 International Conference on Recent Innovation in Science Engineering and*

5. M. Anand, P. Venugopal, D. Al-Tarazi, **G. K. M.**, R. Adakane and P. Subramaniam, "Innovative Optimal Path Selection with Street-Centric Routing in Urban Vehicular Ad hoc Networks," 2025 International Conference on Frontier Technologies and Solutions (ICFTS), Chennai, India, 2025, pp. 1-6, doi: 10.1109/ICFTS62006.2025.11031546. **(IEEE Conference) (Scopus Indexed)**.
6. Ganesh Kumar et. Al. (2024). "Aegis Chain: A Secure and Scalable Digital Wallet Framework for IoT and Cyber-Physical Systems in the Web 3.0 Paradigm". Alliance Doctoral Research And Innovation Symposium Adris 2024. **(Accepted)**.
7. Ganesh Kumar et. Al. (2024). Efficient Adaptive Test Frameworks for IoT Sensor Networks: Precision and Fault Tolerance Purposes. Alliance Doctoral Research And Innovation Symposium Adris 2024. **(Accepted)**.
8. Bhavani, K.D., Sireesha, R., Raju, K.V., Ganesh Kumar, M., Naik, R., & Nagedram, S. (2021). Performance Analysis of 5:2 compressor with 58 transistors. Journal of Physics: Conference Series, 1804. **(Scopus Indexed)**.

Patents:

S.No	Category	Title of Patent	Inventor Position	Status
1.	INDIAN Design Patent Application No. 504271-001	AI-BASED IMAGES PROCESSING DEVICE	Inventor-3	Granted 10-07-2026
2.	INDIAN Utility Patent Application No. 202641034024 A	AI-Enabled Vision System for Real-Time Assembly Step Verification and Worker Action Monitoring	Inventor-2	Published 27-03-2026
3.	INDIAN Utility Patent Application No. 202641011318 A	A Wearable Women Safety Device with Multi-Network Emergency Alert and Real- Time Location Tracking	Inventor -1	Published 13-02-2026
4.	INDIAN Design Patent Application No. 477059-001	IOT BASED CYBER SECURITY DATA ANALYTIC DEVICE	Inventor -9	Granted 06-02-2026

5.	INDIAN Design Patent Application No. 483096-001	AI BASED DEVICE FOR SOFTWARE CHECKING	Inventor -9	Granted 16-01-2026
6.	INDIAN Utility Patent Application No. 202541133623 A	A Secure and Intelligent Online Medical Consultation System for Remote and Personalized Healthcare Delivery	Inventor -1	Published 09-01-2026
7.	INDIAN Design Patent- Design Number 475594-001	AI BASED FOOD QUALITY CHECKING DEVICE	Inventor-3	Published 26-12-2025
8.	INDIAN Utility Patent Application No. 202541118944 A	AI-based Personal Fashion Recommendation System and method.	Inventor-9	Published 19-12-2025
9.	INDIAN Utility Patent Application No. 202541113315	A REAL-TIME COLLABORATION AND KNOWLEDGE MANAGEMENT PLATFORM	Inventor-5	Published 05-12-2025
10.	INDIAN Utility Patent Application No. 202541072525 A	Process for Preparing an Ice Crystal Composite from Borophene Oxide and Silicon Dioxide Nanoparticles	Inventor-2	Published 15/08/2025
11.	German Utility Patent- Model No. 20 2025 102 610	Tobacco sorting system with image processing and IoT	Inventor-4	Granted 27.05.2025
12.	German Utility Patent- Model No. 20 2025 102 619	System for automated feeding of shrimp feed using drone technology	Inventor-7	Granted 27.05.2025
13.	INDIAN Utility Patent Application No. 202541016547 A	IoT Based Virtual Reality Headset	Inventor-3	Published 07/03/2025
14.	German Utility Patent- Model No. 20 2024 106 439	System for Continuous Learning in AI-controlled Autonomous Devices	Inventor-5	Granted 23/01/2025
15.	INDIAN Design Patent- Design Number 441097-001	DATA PROCESSING DEVICE FOR CLOUD LOAD BALANCING	Inventor-5	Published 18/12/2024
16.	INDIAN Utility Patent Application No.202441077987 A	Integrated Supervised and Unsupervised Learning for Mall Customer Segmentation	Inventor-4	Published 25/10/2024
17.	INDIAN Design Patent- Design Number 421054-001	Wearable Activity And Lifestyle Tracking Device	Inventor-10	Granted 16/08/2024
18.	INDIAN Utility Patent	Deep Convolutional Neural Network Classifier for Knee		

	Application No. 202341025319	Osteoarthritis.	Inventor-1	Published 05/05/2023
19.	INDIAN Utility Patent Application No. 202341019159	Classification of Knee Osteoarthritis Using Deep Convolutional Neural Network Classifier	Inventor-1	Published 31/03/2023
20.	INDIAN Utility Patent Application No. 202341014176 A	Early Detection of Knee Osteoarthritis Using Meta- Heuristic Approach	Inventor-1	Published 17/03/2023
21.	INDIAN Utility Patent Application No. 202341003996 A	Early Prediction of Knee Osteoarthritis using Ensemble Machine Learning Approach	Inventor-1	Published 10/02/2023

Book Chapters:

1. Authored a chapter titled “Machine Learning-Based Analytics for Premature Rheumatoid Arthritis and Osteoarthritis Detection in Clinical Practices: A Review” in a book titled “Machine Learning Applications for Data Analysis in Healthcare Systems”, published by Apple Academic Press **CRC Press, Taylor & Francis Group**. E-Book ISBN: 9781779643193. (2025).
2. Ganesh Kumar et. Al. (2025). Human-Centered AI: Enhancing Interaction Design through Intelligent Interfaces and Data-Driven Insights. International Conference on Advances in Communications, Computing & Electronic Systems (ACCES-2025).(**Accepted**).
3. Ganesh Kumar et. Al. (2025). Machine Learning-Driven Congestion Control Strategies in Wireless Sensor Networks. International Conference on Advances in Communications, Computing & Electronic Systems (ACCES-2025).(**Accepted**).
4. Ganesh Kumar et. Al. (2025). AI-Powered Audio and Speech Processing for Next-Gen Human-Computer Interaction. International Conference on Advances in Communications, Computing & Electronic Systems (ACCES-2025).(**Accepted**).

Books Published:

1. Authored a book titled “System and Network Security: Cryptography, Intrusion Detection, Malware Defense, Firewalls, and Web Security” Published by Academic Scholar Press in November – 2025. ISBN: 978-93-47844-70-6. (**1st Author**)

2. Authored a book titled "Quantum Computing Theory, Architecture and Practical Implementation" Published by PENCIL BITZ, Coimbatore, TamilNadu, India, in September - 2025. ISBN: 978-93-89911-72-5. **(4th Author)**
3. Authored an E-book titled "IoT Solutions for Rural Development: Advancing Agriculture, Aqua, and Dairy fields with Wireless Sensor Networks", in December 2024. ISBN: 978-93-341-9153-0. **(1st Author).**

Resource Person for FDPs/ Guest Lectures:

1. Resource Person for One Day FDP on "**Research paper writing and publishing in UGC, Scopus and Web of Science indexed journals**" Organized by Department of Mechanical Engineering in Brahmdevdada Mane Polytechnic, in association with Pencilbitz conducted on 4th October 2025.
2. Presented a Guest lecture on the topic "**Internet of Things (IoT) Architecture & Applications in Smart Systems**" for **Phillipine students** on 8th October 2025 at QIS College of Engineering and Technology, Ongole, Andhra Pradesh.
3. Presented a Guest lecture on the topic "**The key elements for research article writing and publishing**" for VIT-AP scholars on 12th November 2025 in online.
4. About to present a webinar as keynote Speaker in **IEEE** weekly webinar series on the topic "**Knee Osteoarthritis Detection and Severity Classification Using Deep Learning Classifiers: Advancing Automated Diagnostic Solutions in Healthcare**" to be held on 22nd November 2025 in Online mode.

Academic Experience:

S.No	Name of the organization	Designation	Department	Date of Joining	Date of Relieving	Experience
1	QIS College of Engineering and Technology	Assistant Professor	AIML	25 th September 2023	Working	2 Years and 10 months

	(Autonomous)					
2	VIT-AP University	Assistant Professor Jr	CSE	1 st August 2021	6 th September 2023	2 Years
3	SVR Engineering College	Assistant Professor	ECE	1 st January 2014	9 th February 2021	7 Years

➤ **Total Teaching Experience: 11 Years and 6 months.**

Research Profile:

Scopus ID: [58099876100](#)

Google Scholar ID : <https://scholar.google.com/citations?user=AbOtOXAAAAAJ&hl=en&authuser=1>

ORCID ID : [0000-0002-3646-4339](#)

Research Gate Link: <https://www.researchgate.net/profile/Ganesh-Kumar-M-3>

Web of Science Researcher ID: HSF-5028-2023

Google Scholar h-index: 03

Citations : 96



Dr. GANESH KUMAR M

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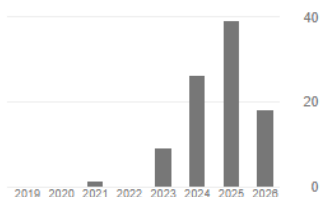
Associate Professor, QIS College of Engineering and Technology (Autonomous)
Verified email at qiscet.edu.in

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Co-authors

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	Agam Das Goswami VIT-AP UNIVERSITY	>
	Udayagiri Prasad ASSISTANT PROFESSOR, JAIN...	>
	Dr. Bhavakar Girish S Ph.D, M.E, B.E.... Associate Professor CSE (AI) Ph...	>
	Dr. Jammisetty Yedukondalu PACE Institute of Technology & ...	>
	MOHAN DAS SARI Assistant Professor, S V R ENGI...	>
	Dr Mohammed Mahaboob Basha Associate Professor, ECE Depart...	>
	Bharathi lakshmanan Professor	>
	Sreenath kocharia Assistant Professor of Informatio...	>
	N Saikiran	>

Professional Memberships:

- ❖ IEEE Member from December 2022.

Achievements:

- ❖ Ratified as an Assistant Professor by JNTU Kakinada in the year 2024.
- ❖ Received Research Award from VIT-AP University in the Year 2023.
- ❖ Received Research Award in Patents category from VIT-AP University in the Year 2023.
- ❖ Received Research Award from VIT-AP University in the Year 2022.
- ❖ Ratified as an Assistant Professor by JNTU Anantapur in the year 2015.

Reviewer:

- ❖ Reviewer for 3rd International Conference on Algorithms and Computational Theory for Engineering Applications (ICACTEA 2026).

- ❖ Third reviewer for the journal titled “International Journal of Computational Intelligence and Applications” (**Scopus Indexed Journal**).
- ❖ Reviewer for IMCEST-2025 – International Multidisciplinary Conference on Engineering, Science & Technology, organized by QISCET in collaboration with Taguig City University & University of Perpetual Help System DALTA (Philippines).
- ❖ Reviewer for third International Conference on Algorithms and Computational Theory for Engineering Applications, **ICACTEA 2K26**, organized by Aditya University, Surampalem.

B.Tech and M.Tech Projects Supervised:

➤ **Project Guide:**

- **B.Tech level 1000+ Students from 2014-till Now**
- **M.Tech 06**

FDPs/Guest Lectures/Co Curricular Activities Organized:

- Organized two days Tech expo of **World Space Week organized by ISRO** from 8th to 9th October 2025 at QIS College of Engineering and Technology, Ongole, Andhra Pradesh.
- Organized one Week FDP from 16th June to 23rd June 2025 on “AI-Driven IoT Systems and Cloud-Enabled Prototyping” at QIS College of Engineering and Technology, Ongole, Andhra Pradesh.
- Organized Poster presentation at QISFEST 2025 from 7th to 9th February 2025 at QIS College of Engineering and Technology, Ongole, Andhra Pradesh.
- Organized one week FDP in hybrid mode on “Emerging Technologies” from 3rd June 2024 to 8th June 2024 at QIS College of Engineering and Technology, Ongole, Andhra Pradesh.

FDPs/Workshops Attended:

- Participated in the Professional Development Programme on “**Pedagogical Strategies for 21st Century Learners**” Conducted by **NITTR** Chennai from 23.06.2025 to 27.06.2025 at QIS College of Engineering and Technology, Ongole, Andhra Pradesh
- Participated in One Week FDP from 16th June to 23rd June 2025 on “AI-Driven IoT Systems

and Cloud-Enabled Prototyping” at QIS College of Engineering and Technology, Ongole, Andhra Pradesh.

- Participated in a Two Week FDP from 1st to 12th April 2024 on “ Recent Advancements in Deep Learning for Computer Vision Applications” conducted by department of ECE, CVR College of Engineering, RR District, Telangana.
- Participated in One Week FDP from 18th to 22nd March 2024, on “ Exploring the Cutting-Edge Developments in VLSI Design” conducted by department of ECE, VNR Vignana Jyothi Institute of Engineering and Technology in association with **IEEE** CAS Student chapter.
- Participated in Five-Day FDP from 26th Feb to 1st March 2024 on “Modeling and Simulation of Electric Vehicles ” conducted by Dhole Patil College of Engineering, Maharashtra.
- Participated in Five-Day FDP on "Designing Edge AI Applications using STMicroelectronics AI Ecosystem" which was held from 14th - 18th of February 2023 organized by VIT-AP University in association with a DigiToad Technologies.

Additional Responsibilities held:

- Working as **Head-Patents and Associate Head- Ideation and Innovation** at QIS College of Engineering and Technology.
- Worked as **Associate Head-Projects** at QISCET.
- Worked as **R&D** coordinator in the department of IT & Allied at QISCET.
- Worked as Co curricular activities coordinator in the department of IT & Allied at QISCET.
- Worked as a Coordinator for building Faculty Profile for National Assessment and Accreditation Council (**NAAC**). Also worked in NAAC **Criteria 1 and 5** at SVREC.
- Worked as an active member for Admission and Fee Regulatory Committee (**AFRC**).
- Worked as a Coordinator for helping the department for **JNTU FFC** at SVREC.
- Worked as an **ALUMNI** Coordinator at SVREC.
- Worked as a coordinator for **VIDYAWAAN** of AP State Government at SVREC.

NPTEL Certificates:

1. Secured an NPTEL Certificate in “**Data Science for Engineers**” with an aggregate score of 57% in a 8-week course conducted from July to September 2025.

2. Secured an NPTEL Certificate in “**Deep Learning**” with an aggregate score of 53% in a 12-week course conducted from January to April 2025.
3. Secured an NPTEL **Elite** Certificate in “**Introduction to Internet of Things**” with an aggregate score of 66% in a 12-week course conducted from July to October 2024.
4. Secured an NPTEL **Elite** Certificate in “**Digital Image Processing** ” with an aggregate score of 68% in a 12-week course conducted from July to October 2019.
5. Secured an NPTEL Certificate in “**CMOS Digital VLSI Design**” with an aggregate score of 40% in a 8-week course conducted from February to April 2019.
6. Secured an NPTEL Certificate in “**Principles of Communication System-I**” with an aggregate score of 42% in a 12-week course conducted from January to April 2018.

Strengths:

Industrious, leadership skills, passionate for teaching, Interactive teaching skills, Good organizing skills, Result oriented, Can counsel and motivate the students.

Personal Details:

Name	: Dr. GANESH KUMAR M
Father's Name	: LAKSHMANA SWAMY M
Date of Birth	: 27/07/1990.
Permanent Address	: H.No:28/1103-A1, Sai Baba Nagar, Nandyal (District), A.P.-518501.
Marital Status	: Married.

Declaration:

I confirm that the information provided by me is true to the best of my knowledge & belief.



Place: Ongole

Your Faithfully
Dr. Ganesh Kumar M.